Taejun Yoon

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Research Interests

Building memory-hierarchy-aware caching systems for LLM inference and cloud analytics

- **LLM Inference Optimization:** Improving the throughput and memory efficiency of LLM serving systems (e.g., vLLM) for long-context and agentic workloads, with a focus on KV-cache management
- **Sparse Attention:** Designing retrieval-based sparse attention with KV-cache offloading to host memory, enabling larger batch sizes for long-context decoding
- **Cloud Data Systems:** Designing and evaluating caching for direct-access cloud OLAP over object storage

Education

POSTECH, Dept. of Computer Science and Engineering — Mar. 2024 – Present
M.S./Ph.D. Integrated Program
Data Systems Lab
Advisor: Prof. Wook-Shin Han

POSTECH, Dept. of Computer Science and Engineering — Feb. 2020 – Feb. 2024
B.S. in Computer Science and Engineering
GPA: 4.07 / 4.3

Professional Experiences

Data Systems Lab, POSTECH, Korea — Jun 2022 – Feb 2024
Research Intern

- Conducted research on database recovery
- Implemented efficient data loading for a distributed database system

Publications

1. **Taejun Yoon**[*], Taesung Lee[*], Joo Yeon Lee, Ki Hong Kim, Byungchul Tak, Wook-Shin Han. ([*]equal contribution) *Cache Replacement in Direct-Access Cloud OLAP—An Experimental Comparison and Practical Guidance [Experiments & Analysis].* SIGMOD 2027.

Projects

Cache Replacement for Direct-Access Cloud OLAP — Mar 2024 – Jun 2026
Industry Collaboration with SAP Labs Korea

- **Problem:** Direct-access cloud OLAP engines rely on local caches to hide remote object-store latency, but existing systems largely inherit recency-oriented policies from DBMS buffer management, ignoring that cloud misses combine a fixed per-request overhead with a size-dependent transfer cost
- **Contribution:** Conducted the first broad experimental comparison of 22 cache replacement algorithms for direct-access cloud OLAP, formulated a latency-aware metric (Unified Hit Ratio), and derived practical policy-selection guidance
- **Impact:** Collected workload traces from live executions of the commercial SAP HANA system over Parquet data on S3, grounding the guidance in a production system
- **Outcome:** Accepted at SIGMOD 2027

Technical Skills

Programming Languages	C/C++ (Advanced), Python (Advanced), SQL (Advanced)
Databases	PostgreSQL, DuckDB
Systems & Build Tools	Linux, Git, Docker, CMake
ML/LLM	PyTorch, vLLM, OpenAI API

Languages

Korean	Native
English	Professional

Teaching Experiences

- **Teaching Assistant**, CSED321: Programming Languages, POSTECH — Spring 2026
- **Teaching Assistant**, CSED421: Database Systems, POSTECH — 2024

References

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